Ze-Yu Xing

PhD Candidate in Physics
Institute of Physics, Chinese Academy of Sciences

xing-phys.github.io/
scholar.google.com/citations?user=qwSu3ekAAAAJ
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Research Profile

My research explores novel phenomena in quantum systems far from equilibrium, particularly in open quantum systems and quantum many-body dynamics. I am interested in the principles governing collective behavior and relaxation, and in how information theory and probability can deepen our understanding of these phenomena. Looking ahead, I aim to explore how AI can advance theoretical discovery and how insights from physics can contribute to quantum computing and new approaches to learning and computation.

Education and Affiliation

Institute of Physics, Chinese Academy of Sciences — Beijing, China
PhD Candidate in Physics — *Aug. 2022 - Present*
Advisor: Haiping Hu.

Jilin University — Jilin, China
B.Sc. in Physics, Honors Program in Science — *Aug. 2018 - Jun. 2022*
GPA: 3.88/4.00 (93.2/100).

Ongoing Research Projects

OYA-OQS: Liouvillian Lieb-Schultz-Mattis Constraints in Open Quantum Systems
Local, translationally invariant Lindbladians with strong U(1) symmetry and a unique steady state in a fixed-charge sector exhibit asymptotically gapless modes in the thermodynamic limit: long-wavelength modes at any filling, plus additional modes at wave vectors set by non-integer filling.

Strong Quantum Mpemba Effect via Spontaneous Symmetry Breaking in Open Systems
Spontaneous symmetry breaking enables the strong quantum Mpemba effect for discrete or continuous, strong or weak symmetries without delicate state engineering. States in the steady-state symmetry sector bypass slow modes and relax faster than closer states overlapping the slow manifold.

Publications and Preprints

[1] **Ze-Yu Xing**[*], Yuncheng Xiong[*], and Haiping Hu — 2026
Geometry-adaptive formulation of non-Bloch bands in arbitrary dimensions and spectral instability
Communications Physics 9, 127 (2026)

[2] **Ze-Yu Xing**, Shu Chen, and Haiping Hu — 2025
Universal spreading dynamics in quasiperiodic non-Hermitian systems
Physical Review B 111, L180203 (2025)

Talks and Presentations

Symmetry-Enforced Strong Quantum Mpemba Effect — 2025
19th Symposium for Young Scholars on Cold Atom Physics, Changsha, China.

Honors and Awards

Director's Scholarship, Institute of Physics, Chinese Academy of Sciences 2025

Ny Tsi-Ze Scholarship, Institute of Physics, Chinese Academy of Sciences 2020

Honorable Mention, Mathematical Contest in Modeling, Consortium for Mathematics and Its Applications 2020

National Scholarship, Ministry of Education, People's Republic of China 2019

Outstanding Student of Jilin University, Jilin University 2019